

User Study

Perception of Virtual Avatar Advisors

Scene-by-Scene Study Flow

Mori (1970)

Foundational theoretical paper — proposes the human affinity curve for robots and artificial figures

Key Findings

The valley: affinity rises with human-likeness, then drops sharply at the near-human threshold, recovering only at complete fidelity

The problem zone: modern production avatars sit at mid-range realism — exactly where the drop is deepest and discomfort is strongest

Movement amplifies: a moving near-human triggers stronger aversion than a static one at the same realism level — critical for animated avatars

What this study adopts

Core motivation for NPR: stylisation deliberately places the avatar below the uncanny zone, trading photorealistic fidelity for affective safety. Directly motivates Research Questions 1 and 3.

Canales et al. (2024)

Virtual doctor avatars at three stylisation levels; video-based trust study with forced-choice advisor selection

Key Findings

Three conditions: realistic, semi-realistic, caricatured virtual doctor — same script, same camera, same voice

Self-report trust: statistically equivalent across all three stylisation levels — no significant differences between conditions

Behavioural selection: semi-realistic attracted the highest rate of “which doctor would you consult?” choices

Dissociation: explicit trust ratings did not predict implicit advisor choice — stylisation affects implicit social judgment

What this study adopts

Entire study paradigm: video-based, scripted advisor, same voice and camera. Trust battery structured around McKnight dimensions. Post-study advisor preference ranking (1st/2nd/3rd) as the behavioural measure — adapted from forced-choice to preserve C2 vs C3 signal under expected C1 dominance.

McKnight & Chervany (2001)

Proposes that trust is not one score but three independent, measurable dimensions

Key Framework

Insight: trust has three independent dimensions (competence, benevolence, integrity), not a single score — rendering may affect each differently

Competence: perceived ability to deliver advice (“does this advisor actually know careers and routes?”)

Benevolence: perceived intent to act in YOUR interest (“is this advisor giving advice that helps me?”)

Integrity: perceived adherence to honest and principled norms (“is this advisor straight with me, even unwelcome advice?”)

What this study adopts

11-item trust battery: 4 competence items + 4 benevolence items + 3 integrity items (Parts E, F, G). Each subscale scored separately. Cronbach's α computed per condition.

Bartneck et al. (2009)

Five validated semantic-differential subscales for human-robot and avatar interaction research

Godspeed Subscales — what each one measures

Anthropomorphism (Part C)
Does the avatar feel like it has a personality, feelings, and intentions? High = feels human. Low = feels like a machine.

Likeability (Part B)
Is the avatar pleasant, warm, and friendly? The key affective dimension — the first thing disrupted when an avatar triggers uncanny valley discomfort.

Animacy *not used in this study*
Does it feel alive and biologically responsive? High = living being; Low = puppet. Not used: animation is held constant across all conditions.

Perceived Intelligence
How competent and knowledgeable the advisor seems. Not used: scripted dialogue is identical across conditions.

Perceived Safety — how non-threatening the avatar feels. N

What this study uses

Part B — Likeability (5 items, 5-point): affective warmth and social appeal of each advisor. **Part C — Anthropomorphism** (5 items, 5-point): tests whether NPR changes perceived human-likeness alongside eeriness, or whether the two dimensions shift independently (H5 mediation path).

Ho & MacDorman (2010)

Validated alternative to Godspeed for uncanny valley research; separates eeriness from human-likeness

Key Findings

The Godspeed problem: existing eeriness items conflated discomfort with human-likeness — two dimensions that can vary independently

Key distinction: an avatar can look human without feeling eerie, or feel eerie while appearing non-human — separate perceptual dimensions

The instrument: 6 semantic-differential items, 5-point, two subscales (eeriness and human-likeness) validated across robotic and virtual stimuli

What this study adopts

Part D — 8 eeriness items, 5-point (verbatim, all 8 original Ho & MacDorman items, both subfactors). Directly tests H1: does NPR reduce eeriness? Eeriness and human-likeness are measured separately to detect if NPR reduces uncanniness without reducing perceived human-likeness.

Alipour et al. (2025)

53 empirical studies at the intersection of uncanny valley effects and trust in human-agent interaction

Critical Findings

Flaw 1 — static images: most studies used static images, not video — invalid for animated avatars whose eeriness is motion-dependent

Flaw 2 — single score: most collapsed trust to one number, masking differential effects across competence, benevolence, and integrity

Flaw 3 — no behavioural measure: self-report alone is insufficient; Canales (2024) showed explicit ratings and implicit choice can diverge

Prescription: valid studies need dynamic video stimuli, multi-component trust battery, and at least one behavioural indicator

What this study adopts

This review directly justifies every design decision: video-based stimuli (not static), 3-subscale McKnight trust battery, and post-study advisor selection as the behavioural indicator.

MacDorman & Ishiguro (2006)

Proposes semantic-differential scales to operationalise eeriness as a quantifiable experimental variable

Key Findings

Category mismatch theory: the UVE is driven not by imperfection in appearance, but by categorical ambiguity — agents positioned between “human” and “non-human” categories produce a cognitive mismatch and a

Semantic-differential scales: proposed measurement instruments for eeriness and strangeness, converting Mori’s qualitative curve into a quantifiable experimental variable used in subsequent validation work

Androids as tools: androids can be positioned at any point along the human-likeness continuum with experimental control, enabling controlled generation of UVE responses on demand

What this study adopts

Original source for the eeriness measurement approach in Part D. The category mismatch account supports the prediction that NPR stylisation reduces eeriness by shifting the Avaturn avatar away from the ambiguous near-human boundary, even without dramatically reducing perceived human-likeness.

McDonnell et al. (2012)

11 rendering styles (ToonPencil to photorealistic); trust, appeal, eeriness, and lie-detection across still and moving stimuli

Key Findings

Non-linear trustworthiness: both the most abstract (toon) and most photorealistic styles received equivalent trust ratings — mid-range styles scored lowest. Directly replicates the valley shape for trustworthiness.

Stylised can exceed photorealistic on comfort: ToonPencil and ToonShaded were rated more reassuring than near-photorealistic HumanHQ2 — highly stylised rendering can surpass photorealistic rendering on perceived comfort.

Motion amplifies the valley: HumanHQ2 became significantly less appealing when moving ($p < .009$) — imperfect motion on a near-realistic avatar is perceptually more costly than on a stylised one, supporting the video stimulus choice.

What this study adopts

Direct empirical precedent for H2 (NPR does not reduce trust relative to photorealistic rendering). Finding that abstract and photorealistic styles receive statistically equivalent trustworthiness is the clearest published evidence. Also supports the choice of video stimuli: motion amplifies the valley for mid-range unappealing styles.

Seymour et al. (2021) — Have We Crossed the Uncanny Valley?

Seymour et al. (2021)

Human-realistic avatar (Digital MIKE) vs caricature avatars; VR headset (n=72) and 2D screen (n≈550)

Key Findings

Human-realistic preferred: Digital MIKE received significantly higher affinity, trustworthiness, and preference ratings than caricature avatars. The second peak is now reachable in real-time rendering

Caricatures = positive engagement: caricature avatars produced no overt negative reactions; participants engaged socially with them — consistent with Mori's first slope where low-realism figures produce positive affect

VR amplifies differences: VR participants showed stronger preference for the human-realistic avatar than 2D screen observers, suggesting immersive display heightens perceptual differences between realism levels

What this study adopts

Grounds H3: both avatar conditions (C2, C3) are expected to produce lower trust than the real person (C1). The C2 vs C3 comparison mirrors Seymour's human-realistic vs caricature comparison, though in this study the NPR shader is applied to the same base avatar rather than an entirely distinct caricature character.

Mayer, Davis & Schoorman (1995)

Defines the ABI model:
Ability, Benevolence, Integrity as three independent trustworthiness factors

The ABI Trust Model

Ability: the cluster of skills and competencies enabling the trustee to have influence in a specific domain. In an advisory context: does the advisor have the knowledge and judgment to provide useful guidance on this particular

Benevolence: the trustee is believed to want to do good for the trustor, apart from self-interested motives. A benevolent advisor looks out for this trustor's specific interests and would not exploit informational asymmetry.

Integrity: the trustee adheres to a set of principles the trustor finds acceptable — honesty, consistency between stated values and actions, and credibility in communication, including disclosing trade-offs even when inconvenient.

What this study adopts

Theoretical foundation for Parts E, F, and G. Ability = Part E (competence and relevant knowledge), Benevolence = Part F (caring and acting in the participant's interest), Integrity = Part G (honesty and transparent reasoning). The forced-choice advisor selection (P1) operationalises the risk-taking behavioural outcome Mayer et al. place downstream of trust.

Bartneck et al. (2009)

Three subscales dropped:
principled reasons why they
do not measure distinct
constructs in this study context

Exclusion Rationale — Animacy, Perceived Intelligence, Perceived Safety

Animacy (Godspeed II) — excluded

All three conditions use the same video recordings with equivalent animation quality. No design manipulation targets perceived aliveness; animation is held constant across C2 and C3. Additionally, Animacy overlaps with Anthropomorphism at the item level (*Artificial/lifelike* appears in both), requiring

Perceived Intelligence (Godspeed IV) — excluded

Covered with greater specificity by the McKnight and Chervany trust subscales in Parts E–G. The Competence items (“This advisor seems knowledgeable about this topic”, “This advisor gave me relevant advice”) are a direct, theory-grounded measure of the same construct, framed in terms of the advisory relationship rather than general cognitive ability. Adding Godspeed IV creates redundancy without additional explanatory power.

Perceived Safety (Godspeed V) — excluded

Measures observer anxiety during physical proximity to a robot. This construct has no meaningful referent in a video-only study where the avatar is not physically co-present. The scale was validated specifically for industrial robot motion scenarios; applied to a seated video advisor it would produce floor-effect ratings with no variance.

Used subscales (Parts B and C)

Part B — Likeability (5 items, 5-pt): affective warmth and social appeal. **Part C — Anthropomorphism** (5 items, 5-pt): perceived human-likeness, enabling the H5 mediation test — whether NPR-driven eeriness reduction is mediated by a shift in perceived humanness.

Nowak & Biocca (2003)

2×3 between-subjects, n=134; agency and visual anthropomorphism effects on telepresence, copresence, social presence

Key Findings

Counterintuitive result: low-anthropomorphic images produced significantly MORE copresence ($F = 6.52, p = .01$) and social presence ($F = 4.90, p = .03$) than high-anthropomorphic images — the high-fidelity **Expectation mismatch:** high-anthropomorphic avatars raise expectations of rich human-level responsiveness. Scripted limited behaviour fails to meet these, reducing the sense of connection. Low-anthropomorphic avatars set **Agency had no effect:** whether participants believed their partner was human or bot did not change social presence. Knowing an agent is automated is not a barrier to meaningful social engagement

What this study adopts

Theoretical support for H2: if the realistic C2 avatar establishes higher perceptual expectations that the video cannot satisfy, the NPR-stylised C3 may perform comparably or better on social engagement. Lower anthropomorphism does not automatically mean lower social presence.

Nowak & Rauh (2005)

n=177; 30 avatar images varying human-likeness and gender cues; measures: anthropomorphism, credibility, homophily, attraction

Key Findings

Non-linear credibility: the relationship between human-likeness and credibility was not linear. Highly anthropomorphic and androgynous avatars received the most favourable credibility and attraction ratings — not the most

More abstract $\neq$ lower credibility: abstract avatars did not automatically produce lower social perceptions. Avatar appearance effects are multidimen-

Appearance reliably affects social perception — human-likeness significantly predicted anthropomorphism, which in turn affected credibility and social attraction — but the pattern was non-linear and moderated by other appearance features

What this study adopts

Establishes that avatar appearance reliably affects social perception dimensions closely related to trust (credibility, attraction). The non-linear pattern supports the expectation that NPR stylisation will affect the perceived credibility of the advisory avatar, though not necessarily in a straightforwardly negative direction.

Weidner et al. (2023)

PRISMA review of 72 empirical papers (2015–2022) on avatars and agents in AR/VR with head-mounted displays

Key Findings

Task-dependent style effects: realistic avatars outperform for bidirectional communication. For game-like and educational tasks, abstract or cartoon avatars perform comparably on task performance and learning outcomes

Rendering style is a meaningful variable: style significantly affects social perception and communication behaviour across dozens of independent studies — confirms NPR stylisation produces measurable perceptual effects

Advisory context gap identified: prior research focused on first-person self-avatars and physical interaction. Third-person advisory trust from scripted video stimuli is an understudied context

What this study adopts

Corroborates the central motivation: NPR stylisation is a meaningful independent variable with measurable perceptual effects. This study fills the advisory context gap Weidner et al. identify, extending avatar stylisation research to third-person advisory trust formation from video stimuli.

Kuffner dos Anjos & Pereira (2024)

n=24; 3 avatar types (Abstract, Mesh, Point Cloud)
× 2 perspectives (1PP, 3PP); 4 VR navigation tasks

Key Findings

3PP amplifies the uncanny valley: realistic Mesh avatar performed worst in 3PP. Realistic appearance sets expectations that imperfect animation fails to meet when observed from outside — motion artefacts are perceptually

Abstract avatars performed better in 3PP: higher body ownership than Mesh in 3PP; fewer spatial errors; greater confidence. Abstract avatars set lower expectations that imperfect motion can meet or exceed

Design guideline: avoid mesh avatars in 3PP when animation fidelity is not guaranteed. Use abstract representations for 3PP tasks where the avatar's full body motion is visible to the observer

What this study adopts

Directly analogous to C2 vs C3: imperfect facial animation on a photorealistic mesh amplifies negative perception when observed from outside. C2 (photorealistic + video) may produce higher eeriness than C3 precisely because subtle animation imperfections are perceptually more costly against a realistic appearance.

Alimardani et al. (2024)

2 × 2 factorial, n=80; appearance × voice (human-like/robotic); emotion recognition task; advice-following rate

Key Findings

Mismatch amplifies eeriness: Humanlike face + Robotic voice produced the highest eeriness, significantly above all matched conditions. The appearance-voice mismatch amplified the uncanny valley response

Critical null — trust behaviour: no significant main or interaction effect of appearance OR voice on advice-following rate. All four conditions produced statistically equivalent trusting behaviour

Clean dissociation: manipulating appearance and voice changes eeriness perception but does not change whether participants follow advice. When demonstrated competence is held constant, perceptual shifts in appearance do not alter trust

What this study adopts

Strongest single-study support for H2. Mirrors the C2 vs C3 comparison: both conditions deliver identical scripted advice. If even a dramatic appearance-voice mismatch does not reduce trusting behaviour, a visual NPR shift on the same advisor is similarly unlikely to produce a significant trust reduction.

Dubosc et al. (2025)

n=40; 3 stylisation levels ×
3 expression intensities; VR
face-to-face; co-presence,
eeriness, trustworthiness

Key Findings

Stylisation reduces eeriness: $F = 11.24, p < .001, \eta_p^2 = .23$. Stylised and semi-realistic produced significantly lower eeriness than realistic ($p < .001, p = .004$). Expression intensity had no effect.

Stylisation increases trustworthiness: $F = 6.94, p < .001, \eta_p^2 = .16$. Stylised and semi-realistic avatars were rated significantly *more* trustworthy than realistic ($p = .001, p = .009$).

Expression content does not confound rendering: the stylisation effect held across all expression intensity conditions — holding the script constant is sufficient to isolate the rendering effect

What this study adopts

Most directly relevant new evidence for H1 and H2. Stylised avatars produce lower eeriness AND higher trustworthiness than photorealistic, suggesting H2 may yield a null or positive direction. Holding the advisor script constant across conditions is sufficient to isolate the rendering effect.

Tao et al. (2025)

Network meta-analysis;
13 studies, 2,343 participants;
3 realism levels;
outcomes: attractiveness,
trustworthiness, eeriness

Key Findings

Eeriness peaks at medium realism: SUCRA 82% for medium realism.

Semi-realistic avatars are the most eerie; both abstract/low and photorealistic/high realism produce less eeriness — first quantitative NMA confirmation

Trustworthiness follows realism monotonically: High (79%) > Medium (44%) > Low (27%). Low-realism avatars score lowest on trustworthiness —

Competing predictions: the countervailing pressure that motivates H2 to C3 reduces eeriness (H1 supported) but the NMA also predicts a trustworthiness cost (countervailing pressure for H2). The two hypotheses together test whether the eeriness benefit can be achieved without the trustworthiness cost.

What this study adopts

Quantitative foundation for the competing hypotheses H1 and H2. The NMA provides the external benchmark: eeriness reduction from C2 to C3 is predicted, but so is a trustworthiness reduction. This study directly tests whether the Dubosc et al. (2025) dissociation holds under advisory framing.

Cihodaru-Ştefanache & Podina (2025)

PRISMA review, 29 studies; embodied conversational agents; outcomes: attractiveness, anthropomorphism, uncanniness

Key Findings

Stylisation is the best UVE avoidance strategy: studies consistently found stylised ECAs produce lower uncanniness than realistic ones. Deliberate low-realism stylisation is the most frequently recommended design.

Voice-appearance consistency is critical: mismatches between visual realism and voice quality are the most consistent cross-study predictor of amplified uncanniness. C3 (stylised appearance + human voice) introduces

Non-verbal timing matters: lip synchronisation accuracy, expression timing, and blink rate all significantly affect uncanniness. Asynchronies between speech and lip movement consistently amplify eeriness

What this study adopts

Validates applying a single NPR shader uniformly to the Avaturn avatar (C3) rather than selectively stylising only certain body regions. The voice-appearance mismatch finding is acknowledged as a limitation: C3 introduces a greater voice-appearance mismatch than C2, and both the eeriness effect and this limitation should be discussed under H1.

[Author et al.] (2024)

Springer *Virtual Reality*;
VR rehabilitation; NPR
conditions: toon, contour,
Gooch warm-cool, watercolor;
movement feedback study

Key Findings

~~Gooch and toon outperform photorealistic for movement feedback:~~

NPR conditions (especially Gooch warm-cool and toon) improved movement correction accuracy versus photorealistic baseline. Contour-only rendering ~~Gooch warm-cool encodes surface orientation as hue:~~ lit surfaces appear warm (yellow-orange), shadowed surfaces appear cool (blue) — a continuous, intuitive orientation cue unavailable in toon banding or edge de-

NPR is a functional design choice, not only aesthetic: NPR stylisation serves perceptual and communicative goals. This supports using stylisation as a principled strategy for avatar design in social VR contexts

What this study adopts

Motivates the V9 Gooch warm-cool shader (V9_GoochWarmCool.shader). Establishes that NPR rendering choices can be grounded in functional feedback theory, not only in aesthetics. Provides the academic basis for including Gooch-style rendering as an NPR technique class alongside toon and edge-based methods.

[MS Research Authors] (2025) — Real vs. Cartoon Avatar Faces in MR: Longitudinal

[MS Research Authors] (2025)

IJHCS (Elsevier); Microsoft Research; longitudinal MR study; real vs. cartoon avatar faces; social presence, eeriness, trust

Key Findings

Cartoon faces reach trust parity by session 3: photorealistic faces held a higher initial trust rating at session 1, but cartoon-stylised faces achieved statistically equivalent trust by session 3 as the uncanny valley response to **Cartoon faces show no uncanny valley response at any session:** eeriness scores for cartoon condition were low and stable across all sessions.

Photorealistic condition showed highest eeriness in sessions 1–2, partially. **Behavioural engagement favoured cartoon faces in sessions 2–4:** willingness to return for a subsequent MR interaction was higher in the cartoon condition from session 2 onward, suggesting a preference reversal driven by accumulated eeriness.

What this study adopts

Contextualises the single-session design of this thesis: the photorealistic trust advantage observed here (H3) may reflect only a first-session effect. Trust parity between C2 and C3 (H2 null result) is consistent with the longitudinal trajectory. Confirms that cartoon stylisation avoids uncanny valley across all sessions, strengthening the expected direction for H1.